

Plan for ECE 399

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1 Objective

Compare magnetic field measuring methods by accuracy, ease of use, and versatility defined by suitability for AC fields of varying frequencies, broad range magnetic field data, and field scanning volume. This could be quantified through graphs/tables/etc. made in excel, via python code, and if necessary through learning other methods such as R programming.

2 Minimum Goals/Expected Outcomes

- 10 paragraph semi-structured literature review of magnetic-sensing equipment
- Line graph of the magnetic field with respect to the position of the magnetometer with data confirmed from published results, simple COMSOL or ELMER FEM simulation, or other equipment like spectrometers
- Understanding how spectrometers can be operated, calibrated and how they can measure magnetic fields of different magnitudes
- Prepare slideshow presentation of findings as a "final exam"

3 Tools

- COMSOL software - Dr. Kim lab
- Spectroscopy equipment and any optical sensing tools (fiber optics if needed)
- Blender software for visualizations - easily accessible
- 3D Printing equipment - The Cube
- HPC data integration if needed

4 Topics for Discussion

- How often will work be done in the lab versus independently
- Research questions: What magnets should be used (see slideshow), specific types of spectroscopy needed, etc.
- Resources on or off campus for learning, fabrication of goods, etc. such as workshops, webinars, and educational databases